System Selection & Problem Statement Formulation

1. **Target System Selection**
    1. https://www.zara.com/us/
    2. Problem Statement & Context of Use: Write a concise problem statement detailing:
        - The platform name is Zara USA. The target user demographics is mainly younger generations as in terms of style. So, the millennial generation would be the main culprit because of the price point (Zara is more expensive than average) but some well off with money gen z generation would also be the main target as well. The core user tasks would involve the ease, or lack thereof, of browsing the site, filtering through desired styles, adding apparel to shopping cart, and attempting to complete a digital shopping cart experience.
        - There are several consequences if the site has any type of interface failure, high task completion time, user frustration, etc. Some of which are the most hurtful to any business is loss of customers. This would not be because of the style or selection of apparel but more so because there is not an ease of use to obtain this apparel. Issues with accessibility can deter customers from buying apparel, concentrating on buying more apparel, or suggesting to friends and family to buy from the same company or not.

Qualitative Contextual Research

Gather empirical qualitative data to understand how real users experience the system:

- **User Research Protocol**
    - Thank you for your time and support in assisting me with this usability study. Let's take a look at the Zara USA website. Zara is a very popular apparel brand for today’s younger generation and I am going to ask you to complete a couple of normal digital shopping tasks. Please think out loud as you are navigating so that I understand what you expect, want, moments of unexpectedness or confusion, etc. There is no right or wrong way of doing this

because this is about the website's overall design and not particularly you.

- **Data Collection & Synthesis**
    - Summary: I was able to get two participants to participate in my Zara digital shopping experience. Both experiences were entirely different but had many similar accessibility pitfalls. It is seen that Zara has many accessibility and usability flaws not only for millennials but gen x as well. I chose to target the targeted audience and the not so targeted audience since this website is available for all people and should be universally designed for everyone to use. I tasked both of them with just finding an outfit they would want from Zara. One subject navigated the men's section and attempted to purchase 3 items of apparel while the other subject navigated the women's section and also attempted to purchase 3 items of apparel.
    - Pain Points: Several pain points to discuss between the two participants.
        - The overload cognitive overload of the landing page: Both participants voiced their opinions on how busy and loud that landing page is. This created issues from the start on navigating the website.
        - Finding the navigation menu: The navigation menu is a hamburger menu in the top left hand corner but neither participant was able to find it because it is 3 lines that blend in with the busy and loud landing page.
        - Product Measurements: Both participants had issues finding their correct size. The product measurement link was written in plain text and not very noticeable. Then the actual menu was very small, 2 columns at a time, and required playing around with it to find out how it worked. No tool tips were present or any help at all.
        - Shipping to store: One participant had issues trying to find a nearby store in order to ship to store. He entered his zipcode and it showed no results. But when he entered the exact zipcode the store was in, then it popped up.
        - Hidden "Continue" button for checkout: In both participants' experience, they both struggled finding the continue button in order to check out once they established if they wanted it shipped to the store or their residence. One was able to

eventually find it while the other could not find it and chose to go elsewhere.

- Direct User Quotes:
    - Landing page: “Very loud and in your face. Very hard to find the navigation bar.” “Woah! Okay where do I click?”
    - Mix up in autopicking gender clothes: “I’m not a man”.
    - Choosing apparel color: “I couldn’t change the color on the previous screen even though the options were shown (on the search results)”.
    - Product Measurements: “It was hard to find since everything is in plain text. There is no welcoming interface here. Just words on a page.” “Only being able to see 2 sizes at a time makes it hard to decide what my size is”. “It was hidden!”
    - Checking out: “How do I check out?”
    - Finding a store nearby: “I know there is a store in Murray. Why isn’t it pulling up?”
    - Finding the last continue button before finalizing checkout: “There it is!” “It’s not worth it. I will order from somewhere else then.”
- Mental Model Mismatches:
    - Landing Page:
        - Users expect a welcoming and highly accessible and usable landing page because this is your landing spot. This is the ultimate introduction to your website and sets the tone for the rest of the website experience.
        - The reality of the Zara website is the landing page was “in your face” and had too much cognitive load for the average person to handle. This can start as an overwhelming experience.
    - Navigation Menu:
        - This is the place users expect to be their navigator the entire time they are on the website. So, the menu must be accessible and available at all times.
        - The reality of the Zara website is that the navigation menu is hidden. It does not stand out and often can be missed. This creates uneasiness while browsing.
    - Product Measurements Table:

        - Users should expect an easy way to compare all of the sizes available that way they can see what best fits them fast and easy.
        - The reality of the Zara website is that the measurements table only shows part of the table and the scroll bar does not work. It is only holding down the mouse to scroll, which is not good accessibility for a desktop website.
    - Shipping to the store:
        - Users should expect when searching for a nearby store that the results will give you the nearest store. And if there are restrictions on distance then there should be a modern distance filter available.
        - The reality of the Zara website is that if you do not insert the exact zip code where a Zara store exists, then it will show no results. But some customers do not know where the nearest Zara store is or what their zipcode is so this gives these types of customers trouble.
    - Checkout experience:
        - Users expect a checkout experience to be easy and quick and painless. Merchants do not want customers to have a lot of time to think about if they should or should not make a purchase. In fact, merchants love customers who shop at spur of the moment.
        - The reality of Zara's checkout experience is it accidentally hides next/continue buttons and this creates frustration and rethinking points for the customer to decide if it's worth it or not to make the purchase.
- Task Breakdown Points:
    - The size chart verification issue:
        - When users are trying to check their product dimensions or size of their apparel to make sure they are getting the right product for them while in the shopping flow, any moment of wrestling with the accessibility or a non responsive scroll bar gets in the way of this flow.
    - Nearby store search:

        - It becomes cumbersome for users when they try to search for a nearby store but cannot get any results unless they enter the exact zipcode. This could trigger losing customers because a lot of customers like to go to the fashion stores to try on clothes to make their final selections.
    - The checkout experience:
        - When users are in the middle or ready to check out, they are usually finished looking around and are ready to spend money. But any walls or issues during this process could trigger uncertainty and untrustworthy vibes for a user to complete their transaction.

Heuristic Evaluation & Accessibility Audit

Conduct a systematic expert review using established HCI heuristics and web standards:

1. **Nielsen's 10 Usability Heuristics Audit (**resource link**)**
    - I performed a systematic walkthrough of the Zara website desktop application. I evaluated each step of the user's experience against Neilsen's 10 Usability Heuristics.
        - Landing Page:
            a. Heuristic Violation: Aesthetic and Minimalist Design
            b. Severity Rating: 3
        - Navigation Menu:
            a. Heuristic Violation: Consistency and Standards
            b. Severity Rating: 3
        - Product Measurement Table:
            a. Heuristic Violation: User Control and Freedom
            b. Severity Rating: 3
        - Nearby Store Search:
            a. Heuristic Violation: Help Users Recognize, Diagnose, and Recover from Errors
            b. Severity Rating: 4
        - Checkout Experience:
            a. Heuristic Violation: Visibility of System Status and Error Prevention
            b. Severity Rating: 4
2. **Baseline Accessibility (WCAG 2.1 AA) Audit**

- Screen the system for critical accessibility barriers using automated browser extensions (e.g., axe DevTools, WAVE) and manual checks. Focus specifically on:
    - Visual color contrast ratios (WCAG SC 1.4.3):
        a. The issue here is Zara's approach to this website is a minimalist aesthetic but it relies on a pale gray and not bolded typography. And this typography is layered with flashy photographic backgrounds that can easily hide the plan text.
        b. My audit finding is using automated checks that can flag instances where the secondary colors like gray would fail the contrast ratio requirement of 4.5:1.
    - Keyboard navigation accessibility and visible focus indicators (WCAG SC 2.4.7):
        a. The issue here is users that navigate via keyboard can face barriers because the styling of the website fades or hides the screen reader or keyboard navigation.
        b. My audit finding is that the Zara website lacks breadcrumbs so that the mouse/keyboard users are forced to use the browser's back button.
    - Semantic document structure and aria-* attribute support for screen reader compatibility.
        a. The issue here is the heavy layout of layout wrappers. These layouts often require robust semantic HTML and assistive markup.
        b. My audit finding is that multiple image tiles lack descriptive aria-label attributes, tool tips, etc. And this can cause screen readers to read generic image tags instead.

Post-Mortem Reflection (Metacognition)

Include written responses to the following two reflection questions in the final section of your report:

1. How did combining qualitative user observations with an expert heuristic evaluation reveal usability flaws that a developer testing their own code would likely overlook?
    i. Typically when a software developer creates an application, they get on a "Happy Path" because they already know the ins and outs

of the websites. Usually, this can create a bias to hidden or even obvious mistakes in the application and often can get easily overlooked. When mistakes like this get overlooked Heuristic Audits can catch the structural, systematic violations when measuring against design principles. Also, qualitative user testing can reveal issues that a developer can often overlook. For example, when searching the nearby store, the developer already knows a zipcode that works so they will search that up to make sure it works but not one that does not work.

2. Identify one heuristic violation that directly creates a legal or ethical accessibility barrier under WCAG 2.1 AA, and explain how resolving that violation improves the user experience for both disabled users and the general public.
    i. The violation that the Zara website has is the failure of visual color contrast ratios because Zara chose minimalist text layers and pale secondary typography. Because by WCAG 2.1 AA standards Zara is required to maintain a minimum contrast ratio of 4.5:1. This creates an issue for visually impaired individuals and this discriminates against them as a whole. The universal resolution is by darkening the font weights or introducing an accessible background overlays that brings the application back to legal compliance for visually impaired users and all users alike.